

BINAY - Lighting Up The Future



BINAY LED-based HIGH, MEDIUM and LOW Intensity Aviation Obstruction Light Beacons

As per International Civil Aviation Organisation (ICAO) requirements Available in Low Intensity, Medium Intensity and High Intensity versions (as per International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with 5-year/3-year warranties

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick Installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED Intensity degradation over its operating lifetime



THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT

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POWERING LED TECHNOLOGIES WORLDWIDE SINCE 1983

cal Center for Retina (MSCR), along with biomedical engineers from ITESM, Mexico, have developed special software for smartphones that can detect eye diseases like diabetic macular edema. The software relies on the camera of a smartphone to perform tests that could detect any abnormality in the thickness of retina.

Dr Juan Carlos Altamirano Vallejo, medical director at MSCR, says that the technology will not replace the specialist but will help detect the disease early so that it could be treated. The software will help general physicians who may not have in-depth knowledge of ophthalmology for detection of eye diseases, so they could refer the patients to specialists.

Flexible electronics now a reality

Imagine that you could stretch or fold your smartphone, laptop or television like a piece of paper and carry it around in your pocket! Researchers from North Carolina State University, the USA, were able to create a transparent, flexible conductor that may make completely flexible electronics a reality.

Inspired by the flexible design of a spring, researchers first created a 3D polymer mould on a silicon surface. They then coated the mould (shaped like a series of identical, evenly-spaced rectangles) with a layer of aluminium-doped zinc oxide, which served as the conducting material. After that, an elastic polymer was applied to the zinc oxide.

Researchers then removed the silicon and the mould, and what they obtained was a transparent material with series of symmetrical, zinc-oxide ridges on an elastic surface. The resultant material was transparent because both zinc oxide and the polymer are transparent. Ridges of zinc oxide allow the structure to expand and contract, making it stretchable.

Kirigami stretchable batteries for flexible electronics

Arizona State University and China's Jinan University have teamed up to create what could become



An ASU research team has used a variation of origami, called kirigami, as a design template for batteries that can be stretched to more than 150 per cent of their original size and still maintain full functionality (Image courtesy: fullcircle.asu.edu)